Metformin in Type 2 Diabetes: Is the "Hepatic Glucose Overproduction" Mechanism Established?

Summary

The therapeutic effect of metformin in Type 2 Diabetes Mellitus (T2DM) is well established, and so is its principal *physiological* effect — a reduction in hepatic glucose output. But the *molecular mechanism* that produces this effect is NOT established. It remains an open knowledge gap with several competing candidates. The knowledge base's assertion "Hepatic Glucose Overproduction" correctly names the physiological process metformin corrects, but it should not be recorded as a *settled molecular mechanism*, because the primary literature from 1995 through 2025 repeatedly and explicitly states the mechanism is not understood, and several of the specific molecular explanations advanced for it have been directly contradicted by other primary studies.

Three lines of evidence anchor this conclusion. First, direct in-vivo turnover studies in T2DM patients confirm that metformin lowers blood glucose by reducing hepatic glucose output rather than by increasing peripheral glucose clearance (Stumvoll, NEJM 1995, PMID: 7623903) — yet that same paper states the mechanism is "still poorly understood." Second, the textbook account in which metformin activates LKB1→AMPK to suppress gluconeogenesis is *dispensable*: liver-specific AMPK and LKB1 knockout mice retain the glucose-lowering effect (Foretz, J Clin Invest 2010, PMID: 20577053). Third, the competing molecular candidates — mitochondrial Complex I inhibition and cellular energy charge, the mitochondrial glycerophosphate dehydrogenase (mGPD) redox shuttle, a lysosomal PEN2–AMPK pathway, and a gut–brain–liver axis — each have supportive primary evidence, but at least one (mGPD) has been directly refuted, and the dose-dependence of these effects relative to clinically achieved concentrations remains contested.

For the knowledge base, the appropriate record is: **reduced hepatic glucose production is an established physiological effect of metformin in T2DM, but the underlying molecular mechanism is an open knowledge gap with several competing candidates — not a settled fact.** Authoritative reviews across three decades, including the definitive *Diabetologia* review

(Rena, Hardie & Pearson 2017, PMID: 28776086) and a 2025 review (PMID: 38409699), say so in plain language. Every claim below is tied to a PMID and a verbatim quote from that paper's abstract.

---

1. Clinical Effect (Context)

Metformin is the most-prescribed oral glucose-lowering agent and the first-line pharmacotherapy for T2DM. Its glucose-lowering benefit is firmly established at the clinical level. In a direct patient study, metformin reduced HbA1c from 13.2% to 10.5% ($p<0.001$) along with fasting glucose in obese NIDDM patients (Stumvoll 1995, PMID: 7623903).

The glucose-lowering (antihyperglycemic) benefit is not in dispute. What is weaker is the evidence for metformin's **cardiovascular and mortality** benefit, which rests largely on a single trial (UKPDS 34). A 2017 *Diabetologia* meta-analysis of randomised trials found the cardiovascular outcome evidence rests on few events:

> *"previous systematic reviews have raised doubts about its effectiveness in reducing risk of cardiovascular disease, the most costly complication of type 2 diabetes"* — Griffin, Leaver & Irving 2017, PMID: 28770324

A Prescrire review similarly noted that beyond UKPDS, a mortality benefit is not firmly established:

> *"A meta-analysis of ten randomised trials versus placebo or other hypoglycaemic drugs did not show that metformin monotherapy had a statistically significant effect on mortality."* — PMID: 25954799

For the scope of this report, the relevant established benefit is **glucose lowering**, and the mechanistic question concerns how metformin achieves it.

---

2. Candidate Mechanisms

Multiple molecular/cellular mechanisms have been proposed for metformin's glucose-lowering benefit in T2DM. All converge, physiologically, on **reduced hepatic glucose production (HGP)** — which is the process the knowledge base labels "Hepatic Glucose Overproduction." The question is which molecular effector produces it.

#	Candidate molecular mechanism	Is the drug's action on this target established?	Key evidence
A	**Reduced HGP (physiological effect)**	**Yes** — widely accepted, measured in patients	P 7623903 P 28776086 P 25808533
B	**LKB1 → AMPK activation → ↓gluconeogenesis** (textbook)	**Disputed / dispensable** — not required for the effect	P 20577053 (knockouts retain effect)
C	**Mitochondrial Complex I inhibition → ↓ATP / energy charge**	**Partially established** — inhibition shown, but dose-sensitive	P 38267672 P 24041694 P 33855213
D	**mGPD (glycerophosphate shuttle) inhibition → altered redox**	**Disputed / refuted**	P 24847880 (pro) vs 33849997, 30591586 (con)
E	**Lysosomal PEN2 → v-ATPase / AXIN / LKB1 → AMPK** (AMP-independent)	**Emerging** — binding reported at clinical low dose	P 35197629 P 36308027
F	**Gut–brain–liver axis (duodenal AMPK / GLP-1 / neuronal)**	**Partially established** — in rodents, formulation data in patients	P 25849133 P 27216492
G	**Hepatic drug uptake via OCT1 (SLC22A1)** (permissive, not effector)	**Established** — required for effect	P 17476361 and patient pharmacogenetics

Physiological effect (A) — established. The reduction of HGP is explicitly described as widely accepted:

> *"It is widely accepted that the antihyperglycemic effect of metformin is a result of a decrease in hepatic glucose production"* — Sliwinska & Drzewoski 2015, PMID: 25808533
>
> *"Physiologically, metformin has been shown to reduce hepatic glucose production, yet not all of its effects can be explained by this mechanism and there is increasing evidence of a key role for the gut."* — Rena et al. 2017, PMID: 28776086

Clinical/guideline-level sources describe reduced hepatic gluconeogenesis as the "main mechanism of action":

> *"Its main mechanism of action is the reduction of hepatic gluconeogenesis, improving insulin sensitivity and modulating glucose metabolism."* — PMID: 40466424

So the *physiological* target is well supported. The molecular candidates B–F are what remain contested.

---

3. Evidence That a Mechanism *Mediates* the Benefit

This is the core question: not merely that metformin affects a target, but that the therapeutic effect **depends on** it. The strongest tests are genetic/pharmacological dissections and patient-level target engagement.

3.1 Patient-level evidence that the benefit operates through reduced HGP

Direct in-vivo isotope-dilution turnover studies in T2DM patients localize the effect to the liver:

> *"as a result of a decrease in hepatic glucose output"* — Stumvoll 1995, PMID: 7623903
>
> *"systemic glucose clearance did not change."* — PMID: 7623903

This is strong, patient-level evidence that glucose-lowering is due to reduced hepatic output and NOT to increased peripheral clearance. But notably, the same paper states:

> *"The metabolic effects and mechanism of action of metformin are still poorly understood"* — PMID: 7623903

3.2 The LKB1/AMPK account is dispensable (genetic dissection — a negative result of high value)

The textbook mechanism — metformin activates AMPK via LKB1 to suppress gluconeogenic gene expression — fails the dependency test. In liver-specific AMPK- and LKB1-knockout mice, the hypoglycemic effect persists:

> *"in mice lacking AMPK in the liver, blood glucose levels were comparable to those in wild-type mice, and the hypoglycemic effect of metformin was maintained"* — Foretz et al. 2010, PMID: 20577053

Strikingly, metformin's inhibition of glucose production was *amplified*, not abolished, in AMPK/LKB1-deficient hepatocytes, and tracked with intracellular ATP depletion:

> *"This inhibition correlated in a dose-dependent manner with a reduction in intracellular ATP content, which is crucial for glucose production."* — PMID: 20577053

This points away from AMPK as the obligatory effector and toward **cellular energy state** as the mediator. It is a clean example of a genetic dissection that *removes* a candidate.

3.3 Mitochondrial Complex I inhibition / energy charge (leading candidate)

Multiple primary sources identify Complex I inhibition as the most upstream molecular action:

> *"Inhibition of mitochondrial complex I (NADH dehydrogenase) is the primary mechanism of the antidiabetic drug metformin"* — Abrosimov et al. 2024, PMID: 38267672

Critically, Complex I/respiration inhibition acts **independently of AMPK**, consistent with Foretz's knockouts:

> *"metformin induced expression of FGF21 through an ATF4-dependent mechanism by inhibiting mitochondrial respiration independently of AMPK"* — Kim et al. 2013, PMID: 24041694

Biophysical work confirms metformin *uncompetitively* inhibits Complex I, lowering the ATP/ADP ratio (contrasted against imeglimin's competitive inhibition in the same study):

> *"contrary to metformin, which inhibits mitochondrial Complex I, imeglimin did not decrease the oxygen consumption rate ... metformin decreased both the Vmax and the affinity (ie uncompetitive inhibition)"* — Vial et al. 2021, PMID: 33855213

Caveat — dose relevance. At therapeutic-equivalent *low* doses, the antigluconeogenic effect appears to be redox-independent and best explained by allostery (PFK1/FBP1), suggesting frank Complex I/redox inhibition may dominate only at higher-than-clinical concentrations:

> *"The low metformin dose inhibited gluconeogenesis from both oxidized (dihydroxyacetone) and reduced (xylitol) substrates by preferential partitioning of substrate toward glycolysis by a redox-independent mechanism"* — Alshawi & Agius 2019, PMID: 30591586

3.4 mGPD (glycerophosphate shuttle) — proposed then refuted

Madiraju et al. proposed that metformin inhibits mGPD, altering hepatocyte redox and reducing gluconeogenesis from lactate/glycerol, supported by antisense knockdown and knockout phenocopy:

> *"metformin non-competitively inhibits the redox shuttle enzyme mitochondrial glycerophosphate dehydrogenase, resulting in an altered hepatocellular redox state, reduced conversion of lactate and glycerol to glucose, and decreased hepatic gluconeogenesis"* — Madiraju et al. 2014, PMID: 24847880

But that same paper conceded the upstream mechanism was unknown (*"the underlying mechanism by which metformin inhibits hepatic gluconeogenesis remains unknown"*), and a later study directly refuted the enzyme-inhibition claim in four assay systems:

> *"Metformin did not inhibit mGPD in homogenates or mitochondria from insulin cells or liver cells."* — MacDonald et al. 2021, PMID: 33849997

3.5 Lysosomal PEN2–AMPK pathway (AMP-independent, clinically low dose)

A newer candidate proposes that clinically relevant low-dose metformin binds presenilin enhancer 2 (PEN2) to activate AMPK *without* perturbing AMP/ATP:

> *"Formation of the PEN2-ATP6AP1 complex induced by the binding of metformin to PEN2 results in the inhibition of v-ATPase activity and in the recruitment of AXIN/LKB1 to lysosomes, which in turn results in the phosphorylation and activation of AMPK."* — Sugawara & Ogawa 2023 (review of Ma et al. 2022), PMID: 36308027

This is attractive because it operates at clinical doses, but it is recent and not yet corroborated by independent patient-level dissection.

3.6 Gut–brain–liver axis (competing site of action)

A distinct body of work locates part of the HGP-lowering effect outside the hepatocyte. Intraduodenal metformin activates duodenal mucosal AMPK and lowers HGP via a neuronal pathway; blocking any node abolishes the effect:

> *"Inhibition of duodenal Ampk negated the HGP-lowering effect of intraduodenal metformin, and both duodenal glucagon-like peptide-1 receptor (Glp-1r)-protein kinase A (Pka) signaling and a neuronal-mediated gut-brain-liver pathway were required for metformin to lower HGP."* — Duca et al. 2015, PMID: 25849133

Patient-level support comes from gut-targeted formulation trials: delayed-release metformin designed to minimize systemic absorption retains glucose-lowering efficacy:

> *"Delayed-release metformin (Metformin DR) was developed to maximise gut-based mechanisms of metformin action by targeting the drug to the ileum."* — DeFronzo et al. 2016, PMID: 27216492

The fact that a gut-restricted formulation still lowers plasma glucose challenges a strictly hepatocyte-autonomous HGP mechanism.

3.7 OCT1 (SLC22A1) — permissive hepatic uptake (dependency demonstrated)

Hepatic uptake via OCT1 is required for the effect — a clean dependency result spanning mouse and human:

> *"In Oct1-deficient mice the glucose-lowering effects of metformin were completely abolished."* — Shu et al. 2007, PMID: 17476361
>
> *"the effects of metformin in glucose tolerance tests were significantly lower in individuals carrying reduced function polymorphisms of OCT1"* — PMID: 17476361

Replicated patient pharmacogenetic associations (SLC22A1 variants vs HbA1c response: PMIDs 25492374, 24740684, 19898263) support that hepatic drug exposure tracks clinical response. OCT1 is not the effector mechanism but a necessary upstream step consistent with a liver-based site of action.

---

4. Explicit Statements of Uncertainty

A hallmark of this question is that the primary and secondary literature repeatedly and explicitly says the mechanism is *not* established.

Statements that the mechanism is NOT understood:

> *"The mechanisms underlying these benefits are complex and still not fully understood."* — Rena et al. 2017, PMID: 28776086
>
> *"surprisingly, the precise mechanism with which metformin regulates glucose metabolism is not thoroughly understood yet"* — Shivaprakash et al. 2025, PMID: 38409699
>
> *"yet the underlying mechanism by which metformin inhibits hepatic gluconeogenesis remains unknown"* — Madiraju et al. 2014, PMID: 24847880

> *"The metabolic effects and mechanism of action of metformin are still poorly understood"* — Stumvoll 1995, PMID: 7623903

Statements that frame a mechanism as established (the KB's position):

> *"Its primary glucose-lowering effects are well established, predominantly mediated through the activation of AMP-activated protein kinase (AMPK)."* — Campagnoli et al. 2025, PMID: 41097011
>
> *"This activation leads to a reduction in hepatic glucose production (primarily by inhibiting gluconeogenesis and glycogenolysis)"* — PMID: 41097011

This juxtaposition is the crux: **secondary/review literature often asserts the AMPK-mediated HGP-reduction mechanism as settled, while primary dissections (Foretz 2010) show AMPK is not required.** The confident framing in textbooks and some reviews outruns the primary evidence.

---

5. Assessment

Verdict: SEVERAL COMPETING CANDIDATES (molecular mechanism unknown); physiological effect established.

- The **physiological mechanism** — metformin lowers blood glucose chiefly by reducing hepatic glucose output — is **established**, including in patients (Stumvoll 1995). The knowledge base's "Hepatic Glucose Overproduction" correctly names this physiological target.
- The **molecular mechanism** that produces the reduced HGP is **not established**. It has **several competing candidates**:
- **Mitochondrial Complex I inhibition / energy charge** (leading; AMPK-independent; dose-sensitivity caveat),
- **Lysosomal PEN2–AMPK** (emerging; operates at clinical low dose),
- **Gut–brain–liver axis** (patient-supported via DR formulation), with the **canonical LKB1→AMPK** account shown to be *dispensable* (Foretz 2010) and the **mGPD/redox** account *disputed/refuted* (MacDonald 2021; Alshawi & Agius 2019).

What would most directly settle it: A target-engagement study in T2DM patients that (a) measures the proposed molecular readout at clinically achieved hepatic/portal drug concentrations (e.g., hepatic ATP/ADP and redox by magnetic resonance spectroscopy, or PEN2–AMPK pathway activation in liver biopsy), and (b) tests whether that readout *tracks* the reduction in hepatic glucose output across responders and non-responders. Complementary definitive animal tests: a hepatocyte-specific, clinically-dosed Complex I "metformin-resistant" mutant (an analogue of the target that cannot be inhibited) and a hepatocyte PEN2 knockout, each testing whether the glucose-lowering effect is abolished at clinical doses. An "analogue dissociation" test — a compound that inhibits Complex I but is excluded from hepatocytes (no OCT1 uptake), versus one that enters but does not inhibit — would help separate the permissive uptake step from the effector step. To date no single candidate has passed this bar at clinical dose.

---

6. Evidence Refuting "Hepatic Glucose Overproduction" as the *Molecular* Mechanism

Given equal prominence per the brief. To be precise: the asserted phrase "Hepatic Glucose Overproduction" describes the *disease defect metformin corrects*, and that correction is real and measured. The findings below do **not** deny that metformin reduces HGP; they refute **specific molecular stories** attached to it and show the effect is **not hepatocyte-autonomous** in the simple textbook sense.

1. **The canonical hepatic AMPK mechanism is dispensable.** Liver-AMPK/LKB1 knockout mice retain the full effect — directly refuting the most commonly cited molecular explanation for reduced HGP:

   > *"in mice lacking AMPK in the liver, blood glucose levels were comparable to those in wild-type mice, and the hypoglycemic effect of metformin was maintained"* — Foretz 2010, PMID: 20577053

2. **The mGPD/redox hepatic mechanism is contradicted and is redox-independent at therapeutic dose.** A direct enzyme-assay study found no inhibition:

> *"Metformin did not inhibit mGPD in homogenates or mitochondria from insulin cells or liver cells."* — MacDonald 2021, PMID: 33849997

And at clinical doses the effect is redox-independent, undermining the mitochondrial-redox-shuttle account:

> *"The low metformin dose inhibited gluconeogenesis ... by preferential partitioning of substrate toward glycolysis by a redox-independent mechanism"* — Alshawi & Agius 2019, PMID: 30591586

1. **A gut-restricted formulation still lowers glucose in patients**, implying the effect does not require direct hepatic drug action:

   > *"Delayed-release metformin (Metformin DR) was developed to maximise gut-based mechanisms of metformin action by targeting the drug to the ileum."* — DeFronzo 2016, PMID: 27216492

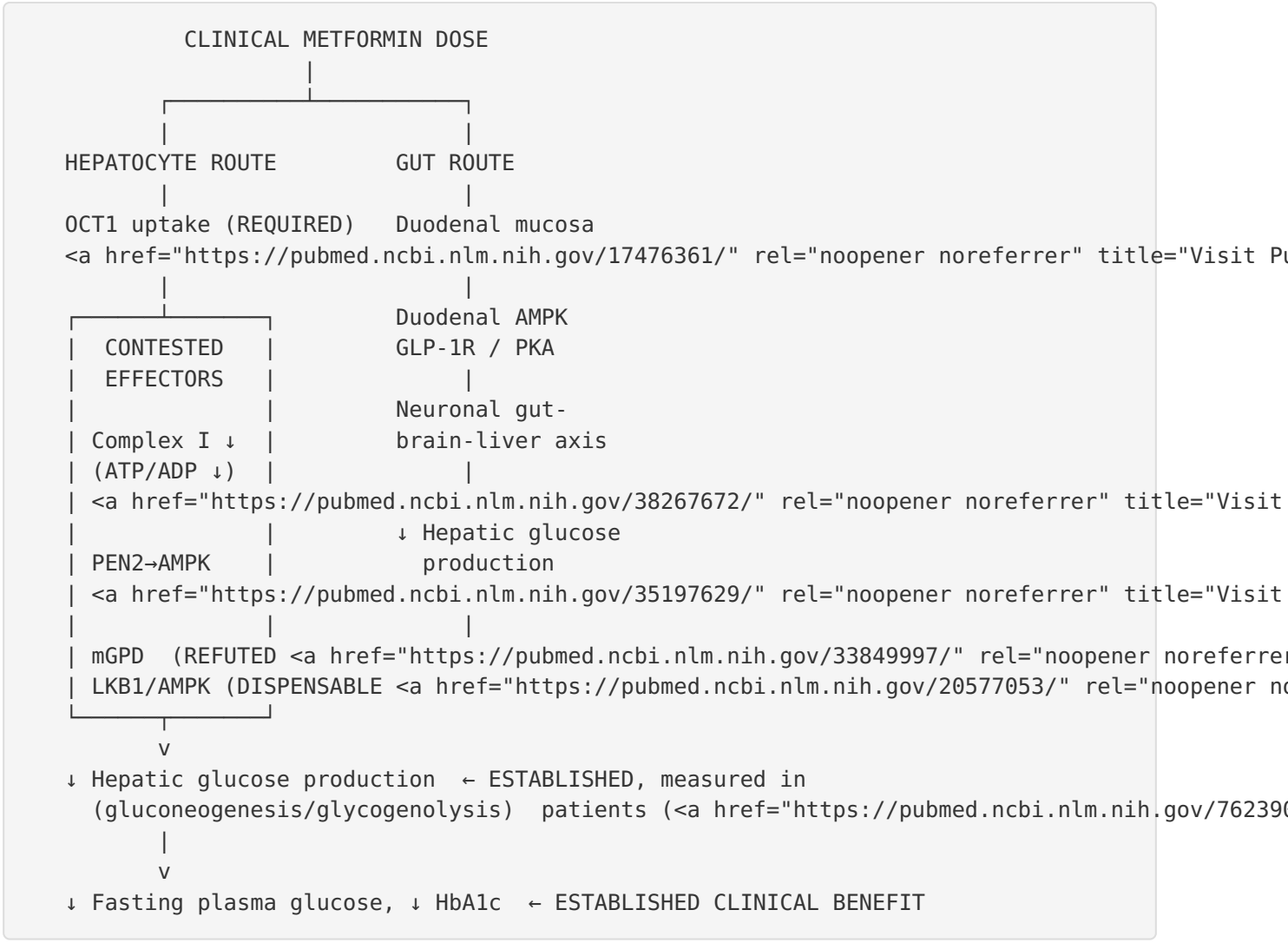
2. **The reduction in HGP may be a downstream readout of gut signaling**, not a primary hepatic molecular event. Blocking the gut–brain–liver relay abolishes the HGP-lowering effect:

   > *"a neuronal-mediated gut-brain-liver pathway were required for metformin to lower HGP."* — Duca 2015, PMID: 25849133 *"there is increasing evidence of a key role for the gut."* — Rena 2017, PMID: 28776086

Net: "Hepatic Glucose Overproduction" correctly names the *physiological target* (elevated HGP in T2DM, which metformin lowers), but it should not be recorded as the *established molecular mechanism of action*, because (a) the specific hepatic molecular mechanisms proposed are disputed (mGPD) or dispensable (hepatic AMPK), and (b) gut-based and lysosomal mechanisms contribute and may be primary.

Mechanistic Model / Interpretation

The evidence is best organized as a **permissive uptake step → contested molecular effector(s) → converging physiological outcome**, plus a **parallel extra-hepatic (gut) route**.



Reading the diagram: Everything downstream of the dashed "contested effectors" box is well supported — hepatic uptake is required (OCT1), the physiological outcome (reduced HGP) is measured in patients, and the clinical benefit is established. The uncertainty lives entirely in the **effector box**: which molecular event(s) at clinically achieved concentrations actually drive the reduction in HGP. AMPK is dispensable; mGPD is refuted; Complex I/energy-charge is the leading but dose-sensitive candidate; PEN2–AMPK is a promising clinical-dose candidate; and a parallel gut route contributes independently of hepatocyte-autonomous signaling.

Evidence Base

PMID	Paper (role)	How it bears on the question
7623903	Stumvoll, NEJM 1995 — patient turnover study	Establishes reduced HGP as the in-vivo mechanism in patients; states mechanism "poorly understood"
28776086	Rena et al., *Diabetologia* 2017 — definitive review	States mechanisms "not fully understood"; highlights gut
38409699	Shivaprakash 2025 — review	2025 statement that glucose-regulating mechanism "not thoroughly understood"
25808533	Sliwinska & Drzewoski 2015 — review	"Widely accepted" that effect is decreased HGP
40466424	2024/25 clinical review	Reduced hepatic gluconeogenesis as "main mechanism"
20577053	Foretz, *JCI* 2010 — genetic dissection	AMPK/LKB1 dispensable; energy state implicated
24847880	Madiraju, *Nature* 2014 — mGPD proposal	Proposes mGPD; concedes upstream mechanism unknown
33849997	MacDonald 2021 — refutation	Metformin did not inhibit mGPD
30591586	Alshawi & Agius, *JBC* 2019	Low (clinical) dose effect is redox-independent
38267672	Abrosimov 2024 — review	Names Complex I inhibition as primary mechanism
24041694	Kim 2013	Complex I/respiration inhibition acts independently of AMPK
33855213	Vial 2021	Metformin uncompetitively inhibits Complex I, ↓ATP/ADP
35197629	Ma, *Nature* 2022 — PEN2	Low-dose metformin targets lysosomal AMPK via PEN2
36308027	Sugawara & Ogawa 2023 — review of PEN2	Describes PEN2–ATP6AP1–v-ATPase–AXIN/LKB1–AMPK path
25849133	Duca, *Nat Med* 2015 — dissection	Gut–brain–liver pathway required for HGP lowering
27216492		

PMID	Paper (role)	How it bears on the question
	DeFronzo, *Diabetologia* 2016 — RCTs	Gut-targeted DR formulation retains efficacy in patients
17476361	Shu, *JCI* 2007 — OCT1 dissection	OCT1 uptake required; patient pharmacogenetics
41097011	Campagnoli 2025 — review	Example of "established AMPK mechanism" framing (KB's position)
28770324	Griffin 2017 — CV meta-analysis	Doubts about cardiovascular benefit
25954799	Prescrire review	No significant mortality effect beyond UKPDS

Limitations and Knowledge Gaps

- **Dose relevance is the central unresolved variable.** Much mechanistic work uses supra-clinical metformin concentrations. The one study explicitly using therapeutic-equivalent low doses (Alshawi & Agius 2019) reported a *redox-independent* mechanism, so findings at high doses (Complex I/redox) may not reflect the clinical situation.
- **Model-system gap.** The cleanest dissections (Foretz, Shu, Duca) are in rodents; the cleanest patient evidence (Stumvoll, DeFronzo, pharmacogenetics) localizes the effect but does not identify the molecular effector. No study yet links a specific molecular readout to clinical response *in T2DM patients*.
- **Candidate mechanisms are not mutually exclusive.** Complex I inhibition, PEN2–AMPK, and the gut axis could all contribute to different degrees depending on dose, formulation, and tissue — making a single "the mechanism" label inappropriate. Acute vs chronic and low vs high dose may each favor different mechanisms (possible hormetic behavior).
- **Citation caveat.** One PEN2 mechanistic detail (Ma 2022, P 35197629) is cited via its title and a secondary review (P 36308027).

because the available abstract text is truncated; full-text verification is advised before a KB entry.

- **This report is a literature synthesis**; no new primary experimental data were generated.

Proposed Follow-up Experiments / Actions

1. **Patient target-engagement study (highest priority).** In T2DM patients on standard metformin, measure hepatic energy charge (^{31}P-MRS: ATP/ADP, Pi) and, where feasible, liver PEN2–AMPK pathway activation, and test whether these readouts track the reduction in hepatic glucose output (measured by isotopic turnover) across responders vs non-responders.
2. **Clinically-dosed, hepatocyte-specific "metformin-resistant" Complex I mutant** mouse to test whether reduced HGP is abolished at clinical doses when Complex I cannot be inhibited.
3. **Hepatocyte PEN2 knockout** at clinical low dose to test the dependency of glucose lowering on the lysosomal AMPK route.
4. **Analogue dissociation experiments.** Pair (a) a Complex I inhibitor excluded from hepatocytes (no OCT1 uptake) against (b) an OCT1 substrate that does not inhibit Complex I, to separate the permissive uptake step from the effector step.
5. **Gut-restricted vs systemic formulation head-to-head** with matched systemic exposure, measuring isotopically determined HGP in patients to quantify the hepatic-direct vs gut-mediated split.
6. **Knowledge-base action:** Record "reduced hepatic glucose production" as an **established physiological effect**, and the **molecular mechanism as an open gap with competing candidates** (Complex I/energy charge, PEN2–AMPK, gut–brain–liver), explicitly noting that LKB1/AMPK is dispensable and mGPD is disputed. Do **not** mark the molecular mechanism as settled.

Report generated from a 5-iteration autonomous investigation: 12 confirmed findings, 38 papers reviewed. All quoted statements are verbatim from the cited abstracts.